

- Matrislerde İşlemler -

(iii) iki Matrisin Çarpımı :

3) iki Matrisin Lineer Kombinasyon Olarak Çarpımı :

Lineer Kombinasyon : $A_1, A_2, \dots, A_n$ aynı büyüklikteki matrisler ve $c_1, c_2, \dots, c_n$ skalarlar olmak üzere

$$c_1 \cdot A_1 + c_2 \cdot A_2 + \dots + c_n \cdot A_n$$

ifadesine $A_1, A_2, \dots, A_n$ nin bir lineer kombinasyonu denir.

3) iki Matrisin Lineer Kombinasyon Olarak Çarpımı :

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \quad \text{ve} \quad x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \quad \text{olmak üzere}$$

$$\begin{aligned} A \cdot x &= \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \end{pmatrix} \\ &= x_1 \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix} + x_2 \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix} + \dots + x_n \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix} \end{aligned}$$

ezeliğinden yavetknp çarpma işleml yapılır.

Örnek:

$$A = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 3 & 0 \\ 0 & 5 & -1 \end{pmatrix} \text{ ve } B = \begin{pmatrix} 4 \\ -2 \\ -1 \end{pmatrix} \text{ için } A.B = \underline{\underline{\begin{pmatrix} 7 \\ -2 \\ -9 \end{pmatrix}}}$$

$$\begin{aligned} A.B &= \begin{pmatrix} 3 & 2 & 1 \\ 1 & 3 & 0 \\ 0 & 5 & -1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ -2 \\ -1 \end{pmatrix} = 4 \cdot \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + (-2) \cdot \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix} + (-1) \cdot \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \\ &= \begin{pmatrix} 12 \\ 4 \\ 0 \end{pmatrix} + \begin{pmatrix} -4 \\ -6 \\ -10 \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 7 \\ -2 \\ -9 \end{pmatrix} \end{aligned}$$

Örnek:

$$A = \begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix} \begin{pmatrix} 2 \\ 6 \\ 0 \end{pmatrix} \text{ ve } B = \begin{pmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \text{ için } A.B = \begin{pmatrix} 1 & 3 & 4 & 0 \\ 4 & 10 & 10 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\left. \begin{aligned} 1 \cdot \begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix} + 0 \cdot \begin{pmatrix} 2 \\ 6 \\ 0 \end{pmatrix} + 0 \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} &= \begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix} \\ 1 \cdot \begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix} + 1 \cdot \begin{pmatrix} 2 \\ 6 \\ 0 \end{pmatrix} + 0 \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} &= \begin{pmatrix} 3 \\ 10 \\ 0 \end{pmatrix} \\ -2 \cdot \begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix} + 3 \cdot \begin{pmatrix} 2 \\ 6 \\ 0 \end{pmatrix} + 1 \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} &= \begin{pmatrix} 4 \\ 10 \\ 0 \end{pmatrix} \\ 0 \cdot \begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix} + 0 \cdot \begin{pmatrix} 2 \\ 6 \\ 0 \end{pmatrix} + 0 \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \end{aligned} \right\} AB = \begin{pmatrix} 1 & 3 & 4 & 0 \\ 4 & 10 & 10 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$